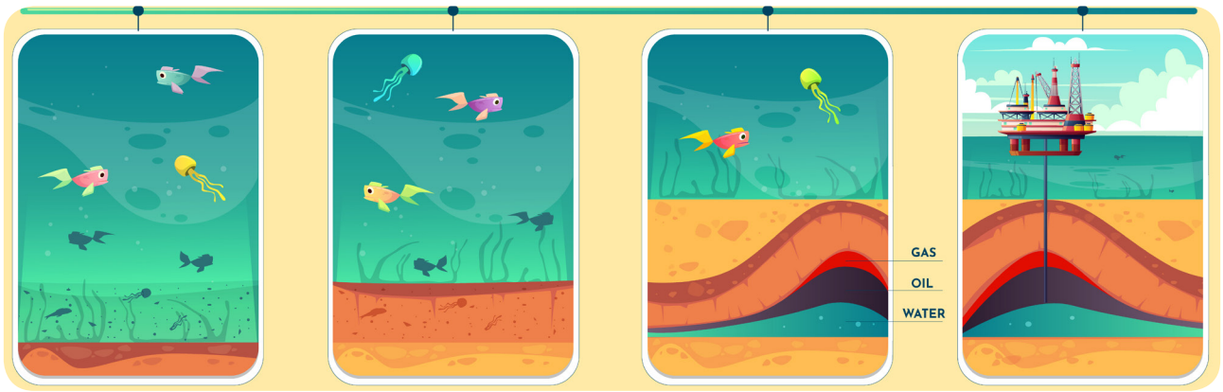



**Figure 15.1** Formation of Hydrocarbons

## 15.1.2 Properties of Hydrocarbons

Among all the chemical compounds hydrocarbons have some unique properties. Some of them are given below.

- Most of the hydrocarbons are insoluble in water.
- Hydrocarbons are less dense than water. So they float on top of water.
- Most hydrocarbons react with oxygen to produce carbon dioxide and water.
- Hydrocarbons can be gases (E.g. methane and propane), liquids (E.g. hexane and benzene) or waxes (paraffin).
- Hydrocarbons are capable of making bonds with one another. This property is known as catenation (chain formation). Due to this property they form more number of complex molecules.

## 15.1.3 Types of Hydrocarbons

In hydrocarbons carbon and hydrogen atoms are linked together through different chemical bonds. Depending on the bond between these atoms there are number of hydrocarbons. The four general classes of hydrocarbons are: alkanes, alkenes, alkynes and arenes. Some of the common hydrocarbons are methane, ethane, propane, butane and pentane.

Methane is the simplest hydrocarbon in which four hydrogen atoms are linked with one carbon atom. It is a colourless, odourless and inflammable gas. It is an eco-friendly fuel because it does not produce any harmful products. It is used as a fuel in electricity generation. Methane is also known as marsh gas as it is present in marshes. Dead and decaying plants and animals release methane gas. It is a renewable source of energy. Sewage sludge can also be decomposed by microorganisms to produce methane gas along with impurities like carbon dioxide and hydrogen sulphide. After removing these impurities, methane gas can be used as an efficient fuel.

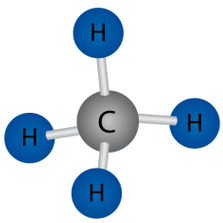
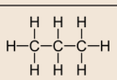
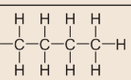
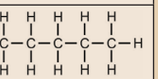



**Figure 15.2** Structure of methane

### Activity 1

Make a model using clay and match sticks for the following hydrocarbons.

| Name | Formula | Structure |
|---|---|---|
| Methane | $CH_4$ |  |
| Ethane | $C_2H_6$ |  |
| Propane | $C_3H_8$ |  |
| Butane | $C_4H_{10}$ |  |
| Pentane | $C_5H_{12}$ |  |



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